part. The thigh bones are attached to either side of the girdle with a ball and socket joint.

c. Upper limb bones

Upper limb bones consist of humerus, radius, ulna, carpals, metacarpals and phalanges. Humerus makes up the upper arm. Fore-arm is made up of radius and ulna. Wrist is made up of carpals. Palm is made up of metacarpals. Fingers are made up of phalanges.

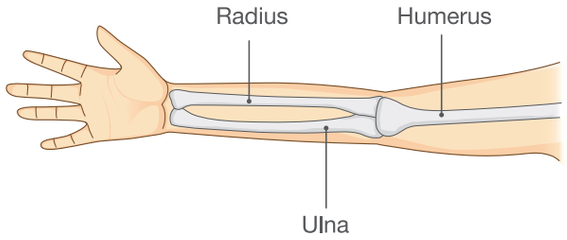


Figure 19.14 Upper limb bones

d. Lower limb bones

Lower limb bones consist of femur, tibia, fibula, tarsals, metatarsals and phalanges. Knee is covered by a cap like structure called as patella or a knee cap. Femur makes up the thigh bone. Leg is made up of tibia and fibula. Tarsal bones are present at the proximal region of the foot. Foot is made up of metatarsals. Toes are made up of phalanges.

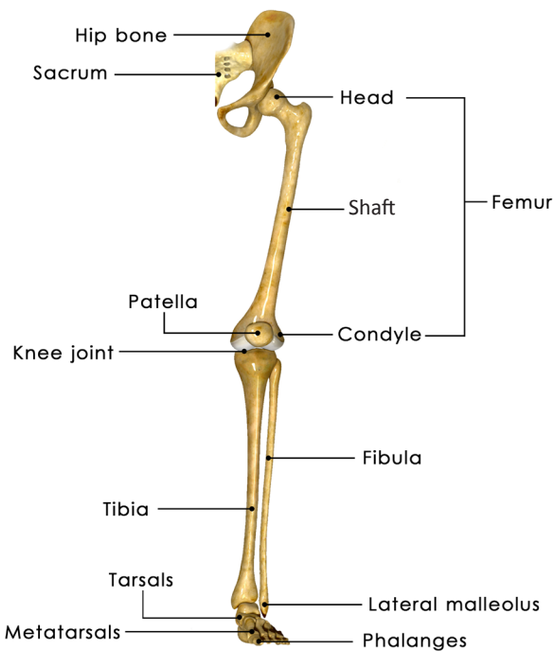


Figure 19.15 Lower limb bones

19.6 Muscles

The muscles in the body provide the means of all movements. They cover the skeletal framework and also give shape to the body. Muscles help to maintain body posture while sitting, standing or walking. Most muscles are long bundles of contractile tissue. Each muscle usually has two ends - **a fixed end** where the muscle originates and **a movable end** which pulls some other part. This **movable end** is drawn out to form a tough structure the **tendon** which is attached to the bone. When stimulated by a nerve the muscle contracts to become shorter and thicker and thus it pulls the bone at the movable end. Muscles can only contract and relax, they cannot lengthen.

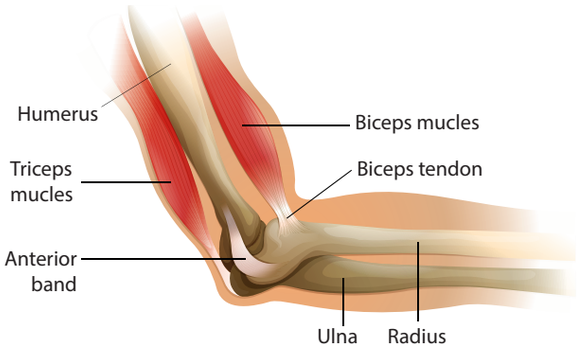


Figure 19.16 Tendon and muscle attachment to the bone in human

Muscles often work in pairs which work against each other. These are called *antagonistic* pairs. The muscles in the upper arm control the bending and straightening of the arm. The two muscles, the biceps and triceps are working

DO YOU KNOW?

- There are muscles in the root of your hair that give you goose bumps.
- It takes 17 muscles to smile and 42 muscles to frown.
- The hardest working muscle is in eye.

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